

CHEESE V4 – COMPLETE SYSTEM ARCHITECTURE & WIRING DIAGRAM

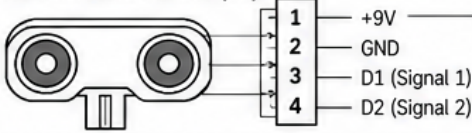
WRO Future Engineers 2026 | Hardware, Communication, Power and Data Flow

Rev. 1.0

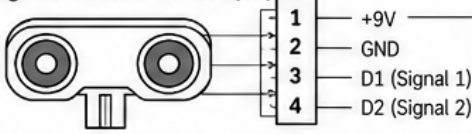
2026-09-08

SENSORS (INPUT)

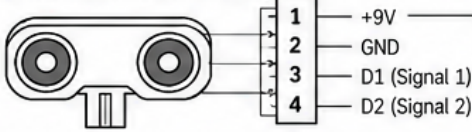
Left Ultrasonic Sensor (S1)



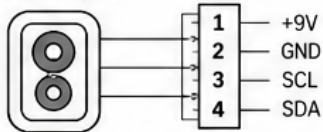
Right Ultrasonic Sensor (S2)



Front Ultrasonic Sensor (S3)

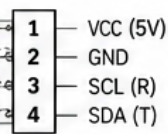
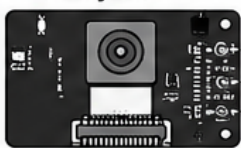


Color Sensor (S4)



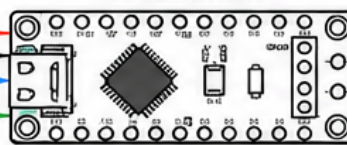
CAMERA SYSTEM

HuskyLens



I²C (Object data:
color & position)

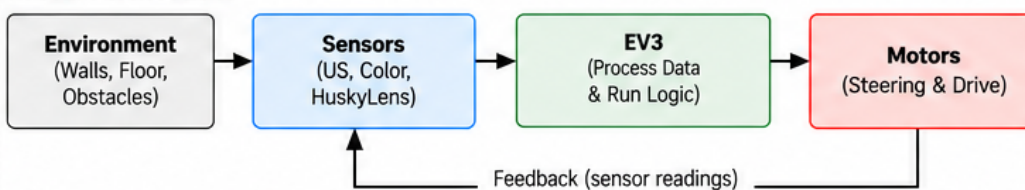
Arduino Nano



WIRE LEGEND

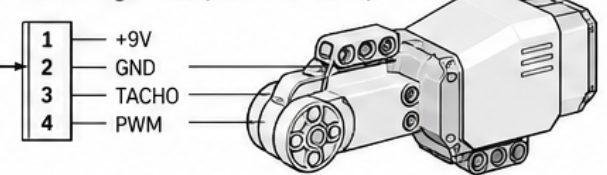
- +5V (from EV3 USB)
- GND (Ground)
- +9V (from EV3 ports)
- Signal (Digital / PWM / Tacho)
- I²C – SCL
- I²C – SDA
- USB (Data + 5V)
- Not Connected

SYSTEM DATA FLOW

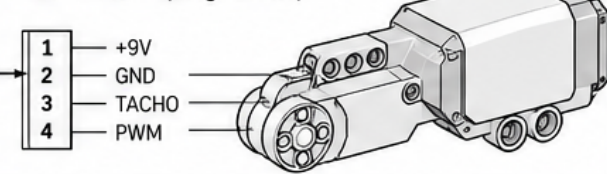


MOTORS (OUTPUT)

A – Steering Motor (Medium Motor)



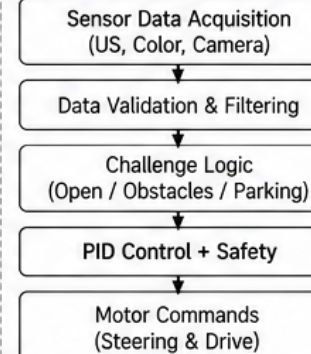
B – Drive Motor (Large Motor)



C – Not used

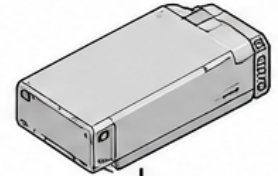
D – Not used

SOFTWARE (EV3)



POWER SYSTEM

EV3 Rechargeable Battery
(7.2 V, 2050 mAh, Li-Ion)



Powers EV3 Brick
(sensors, motors, Arduino Nano
via USB)

NOTES

- EV3 provides 9V to sensors and motors through its ports.
- Arduino Nano is powered via EV3 USB (5V).
- HuskyLens communicates with Arduino Nano using I²C (T = SDA, R = SCL).
- Output ports C and D are not used.
- Wire colors are for reference only.
- This diagram represents Cheese V4 configuration.

Project: Cheese V4

Diagram: Complete System Architecture & Wiring

Version: 1.0

Date: 2026-09-08

Team: Go!Cheese

